

NDMM: Getting Started with Myeloma Management

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Myeloma Basics

Incidence: ~36,000 Americans diagnosed this year.

Median age at diagnosis: 69



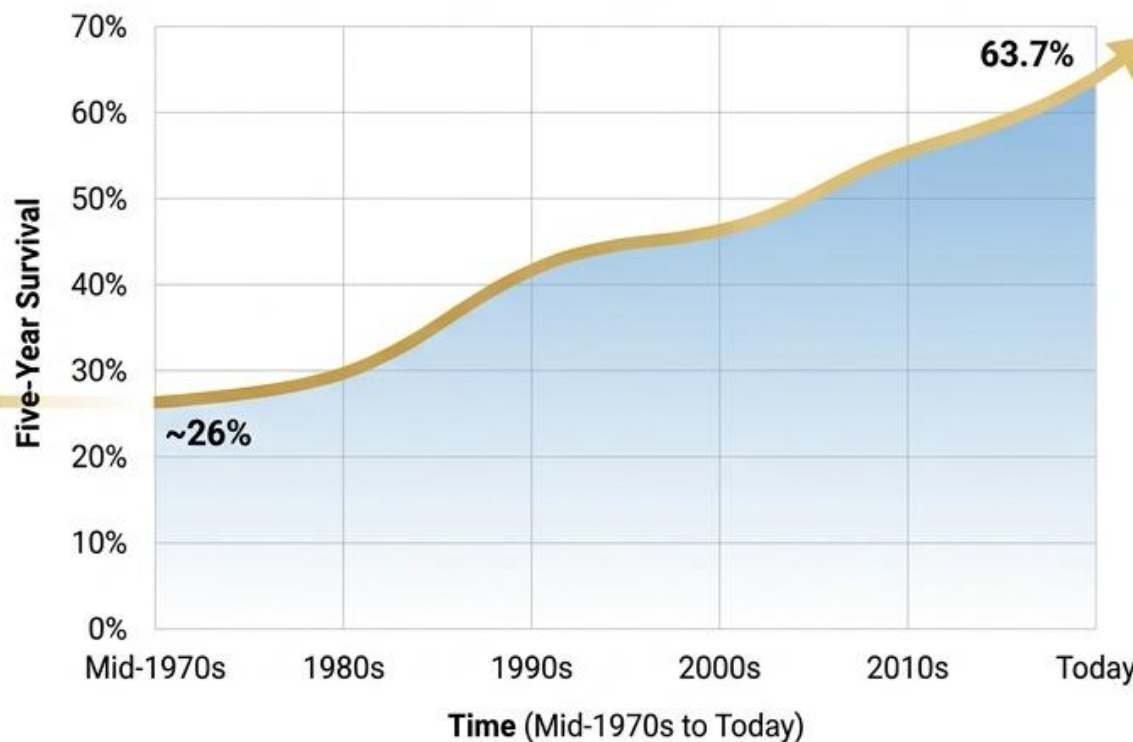
Accounts for just under 2% of all new cancer cases.



The Equity Gap:

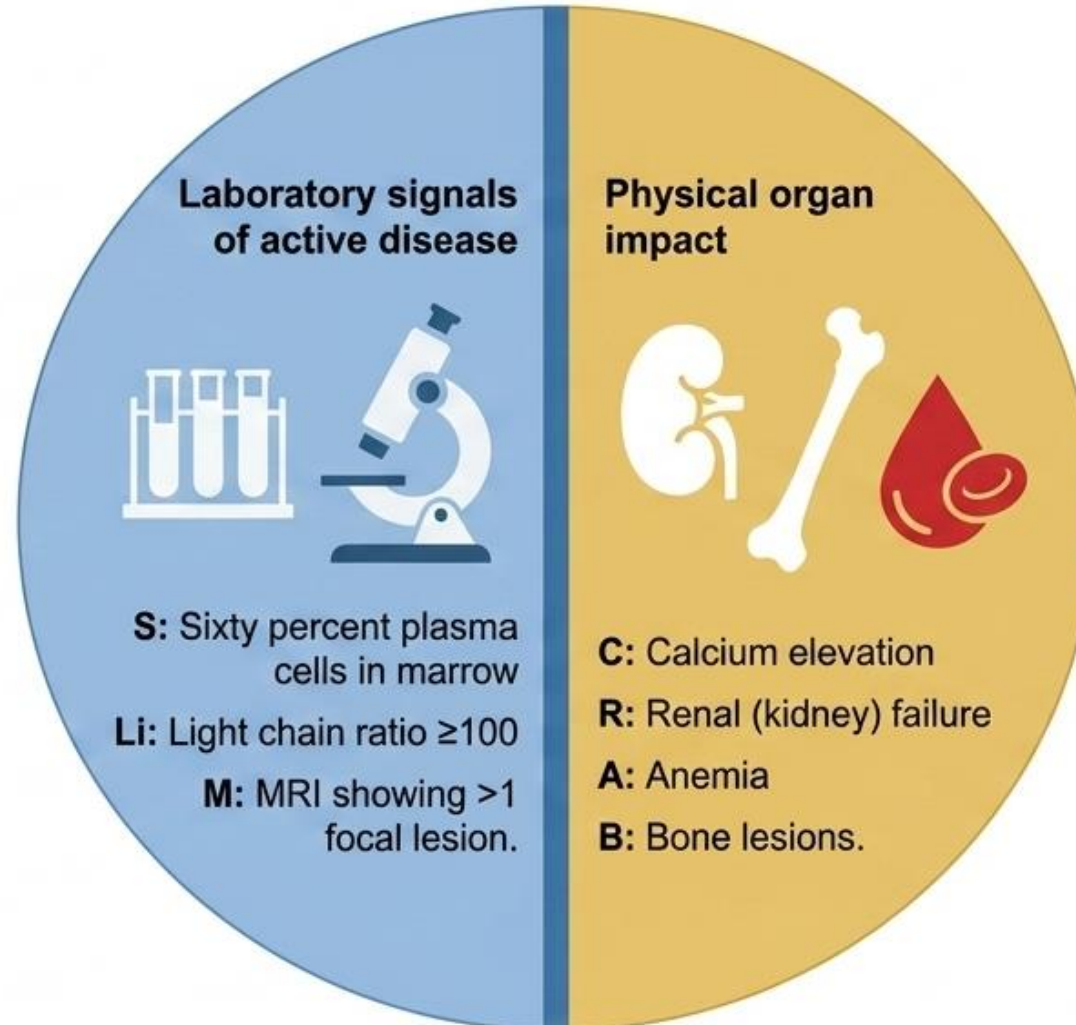
Black Americans are diagnosed at roughly double the rate of white Americans.

Despite this higher burden, Black patients have historically been underrepresented in trials—a gap the field is actively working to close.



This rapid progress is exactly why treatment recommendations continuously evolve.

Defining Active Myeloma: SLiM-CRAB





Baseline Workup

Checklist



Blood protein tests: SPEP, immunofixation,
free light chains (FLC).
24-hour UPEP



Bone marrow biopsy with CD138-selected
FISH testing.
*Crucial Specification: Running genetics on unselected
cells is a common omission*



Full-body imaging like PET-CT or
whole-body MRI.
*Crucial Specification: Standard skeletal X-ray surveys
are no longer sufficient.*

CBC, CMP, LDH, Beta-2-Microglobulin



When Things Move Fast

Certain presentations require care
teams to accelerate the timeline
immediately. These include:

Kidney
failure

High Calcium

Plasma cell
leukemia
(≥5% circulating
clonal cells)

Disease
outside the
marrow

Hyperviscosity
(blood-thickness
emergency)



Ruling Out Look-Alike Conditions

AL Amyloidosis

POEMS Syndrome

Myeloma can resemble two related, rarer conditions: AL amyloidosis (abnormal protein deposits in organs) and POEMS syndrome.

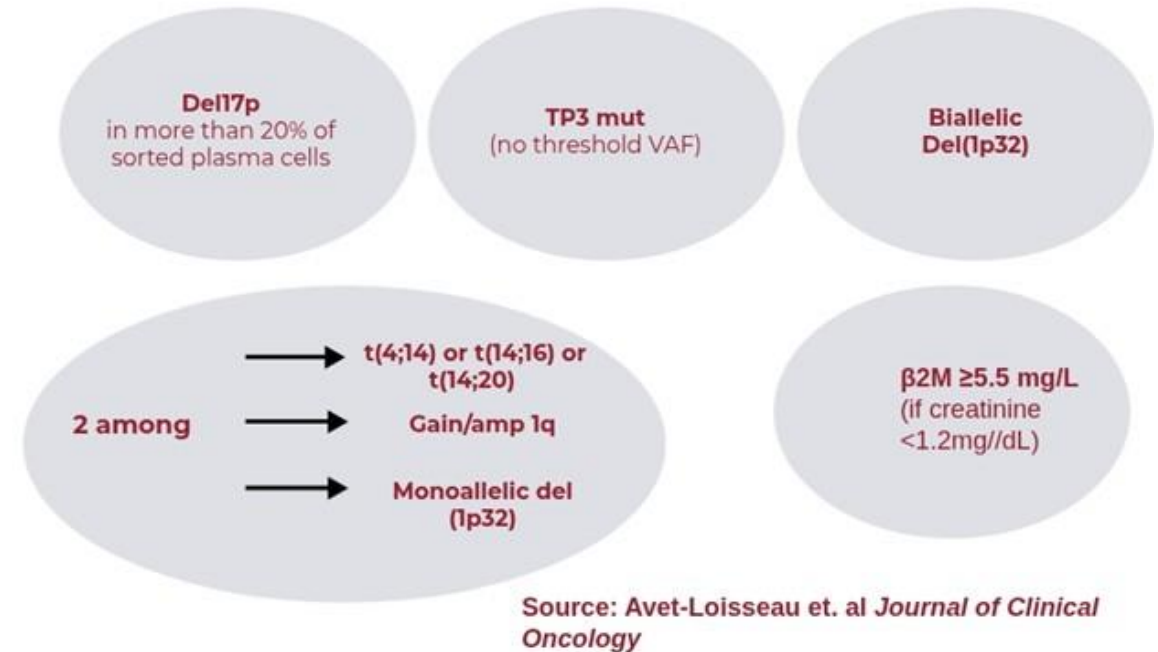
Specific additional testing and clinical findings tells these apart – treatment may differ.

Staging: The Shared Language



R-ISS	Definition (Palumbo et al., JCO 2015)
I	Albumin > 3.5 g/dL, B2M < 3.5 mg/L + standard-risk cytogenetics (FISH) + normal LDH
II	Not R-ISS Stage I or III
III	B2M > 5.5 mg/L + (high-risk cytogenetics OR elevated LDH)

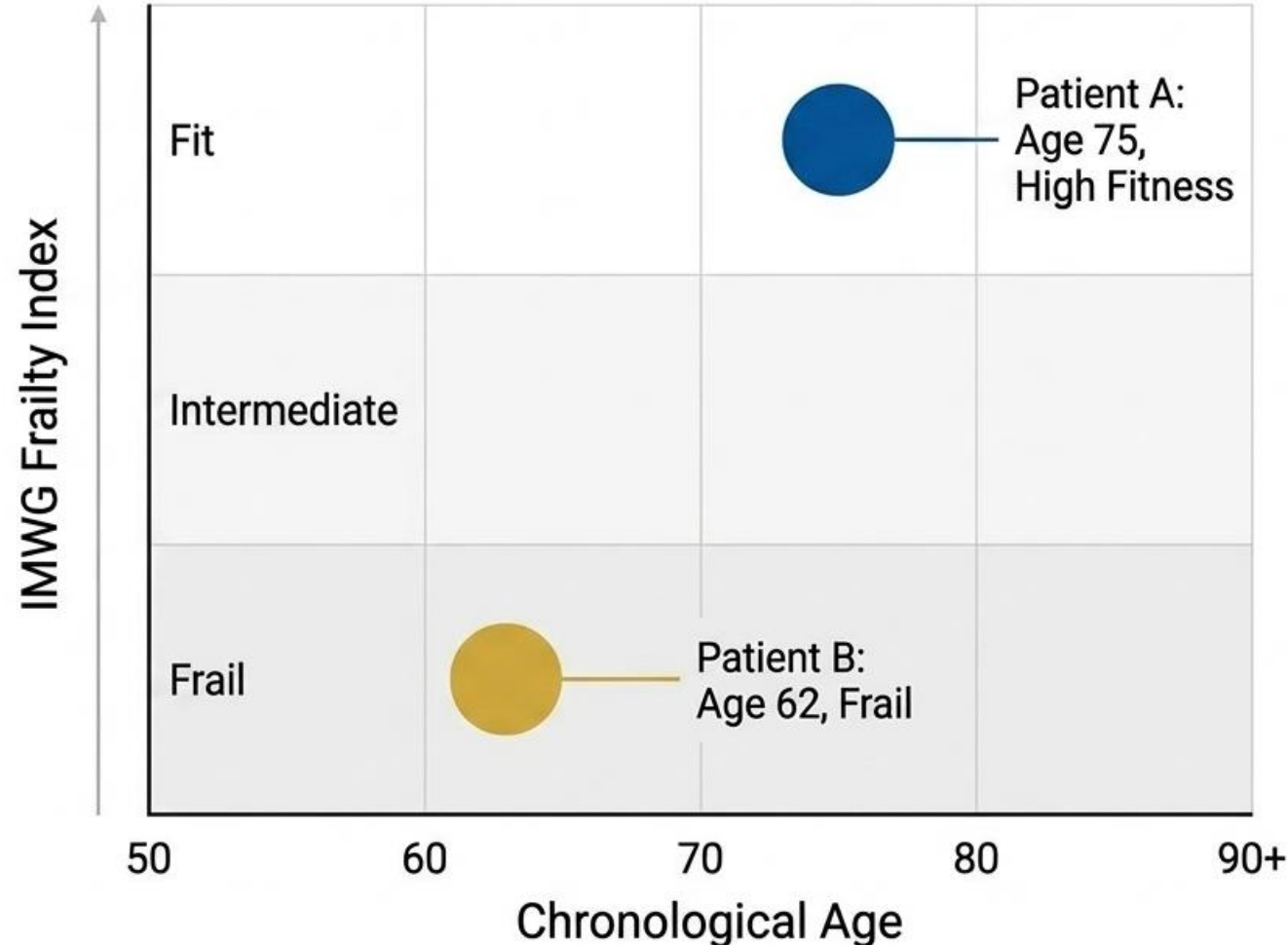
IMS/IMWG consensus on high-risk myeloma definition



R-ISS and IMS/IMWG HR consensus give a shared staging vocabulary from blood tests, and genetics.

Staging is a starting point, not the whole picture.

Evaluate Fitness, Not Age



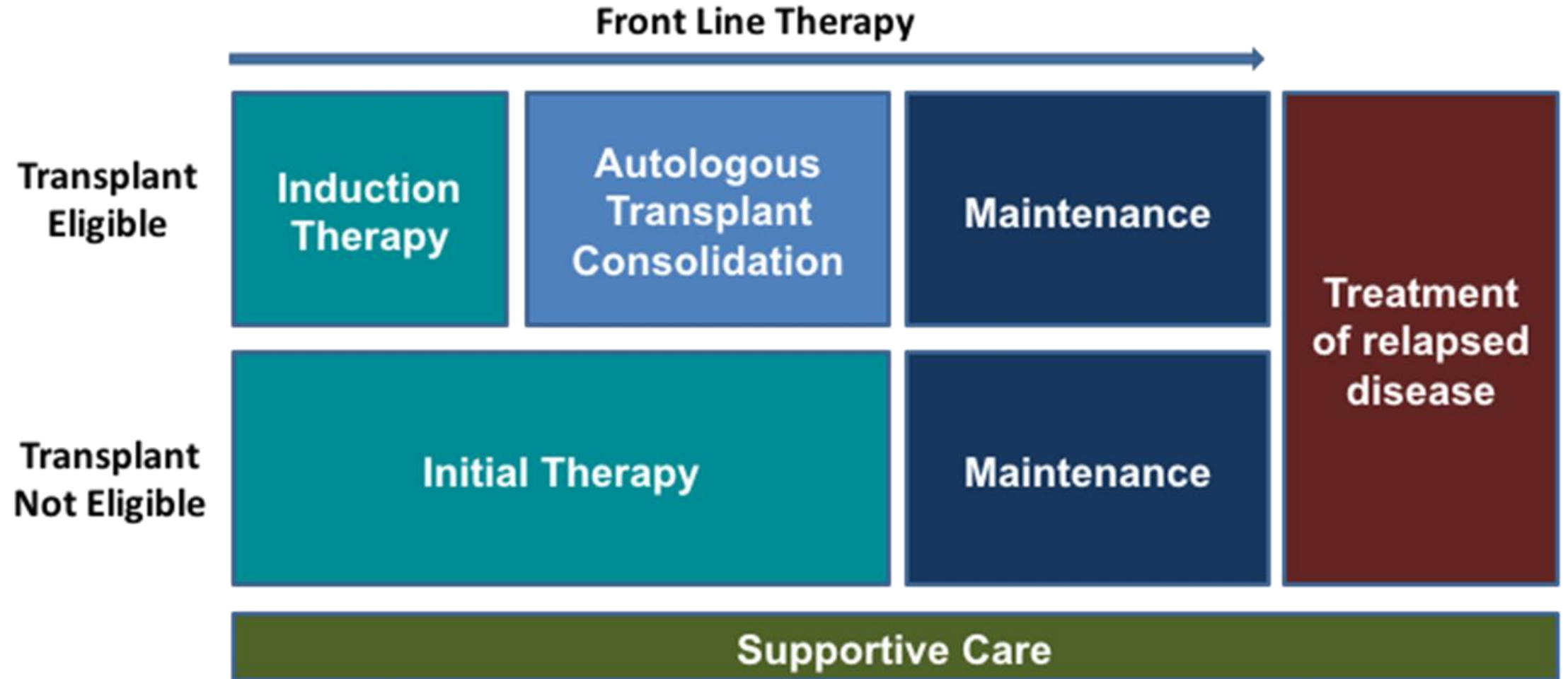
Key Insight:

Frailty is measured by a structured assessment (daily activities, comorbidities), not just the calendar.

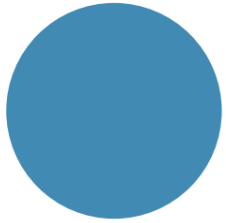
The "Rule of Frailty":

A frailty score modifies the DOSE and SCHEDULE of treatment. It does not dictate which drug classes are on the table. It adjusts intensity without ruling out effective combinations.

The Current Roadmap



The Standard of Care Toolbox



Anti-CD38

Daratumumab
(D: Darzalex)

Isatuximab
(Isa: Sarclisa)

Infusion Reactions
Infections
Fatigue



PI

Bortezomib
(V: Velcade)

Carfilzomib
(K: Kyprolis)

Neuropathy (V)
Heart, Lung or Kidney
(K)



IMiD

Lenalidomide
(R: Revlimid)

Rash
Diarrhea/Constipation
Brain Fog
Blood Clots
Second Cancers



Steroid

Dexamethasone
(d)

Prednisone
(P)

Emotional
Withdrawal



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Am I Transplant Eligible?

Transplant Eligible

Age (?)

Normal organ function (?)

Fit patient

Patient choice

Transplant Ineligible

Age(?)

Comorbidities

Frail patient

Patient choice



Standard of Care in Transplant Ineligible MM

Outcome	MAIA ¹⁻⁵	IFM 2017-03 ⁶
Study regimen	DRd vs Rd (N = 737)	DR(d) vs Rd (N = 295)
Study phase	III	III
Study population	Transplant-ineligible NDMM	Patients aged ≥65 yr with NDMM; IFM frailty score ≥2
ORR, %	93 vs 82; <i>P</i> <.0001	96 vs 85; <i>P</i> = .001
Median PFS, mo	61.9 vs 34.4; <i>P</i> <.0001 Median follow up: 64.5 mo	Not reported
Median OS, mo	90.3 vs 64.1; <i>P</i> <.0001 Median follow up: 7.5 yr	Not reported
MRD at 10 ⁻⁵ by NGS, %	32.1 vs 11.1; <i>P</i> <.0001	10 vs 3; <i>P</i> = .012

Do I BENEFIT from adding a 4th Drug?

Patients aged 65-79 yr with NDMM who are non-frail and transplant-ineligible; no prior systematic treatment; measurable disease; ECOG PS ≤ 2

Cycles 1-12
28-day cycles

Isatuximab + Rd + Bortezomib

Isatuximab + Rd

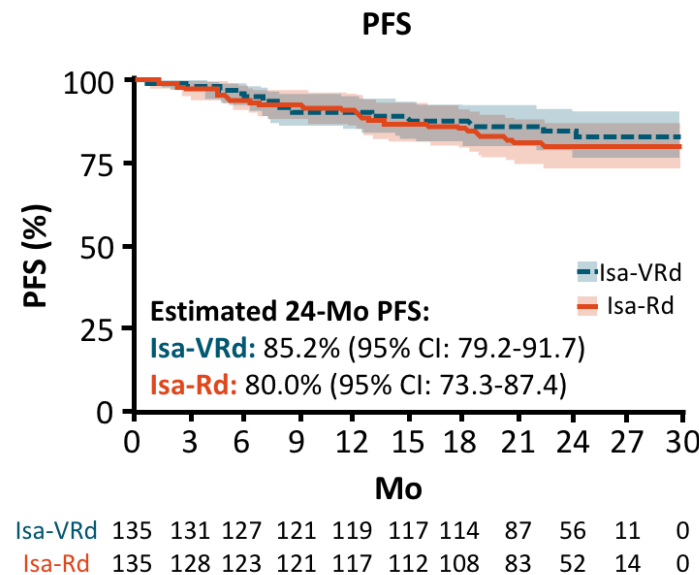
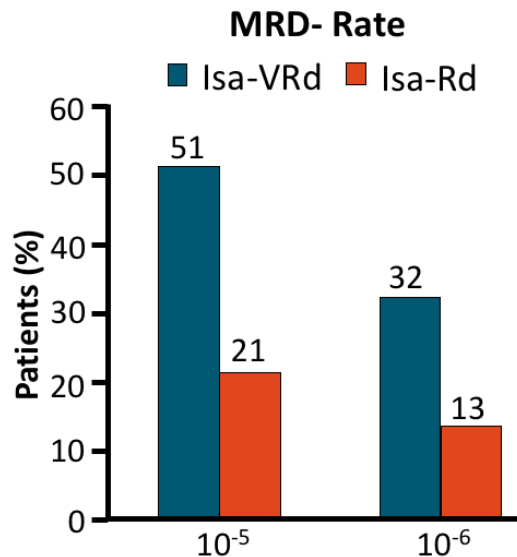
Cycle 13-18
28-day cycles

Isatuximab + R +
Bortezomib*

Isatuximab + R

Cycle 19 onwards
28-day cycles

Isatuximab
+ R



Transplant Ineligible MM – Summary



Outcomes continue to improve with three and four drug combinations!

Dara and Isa combinations appear safe ( infections)

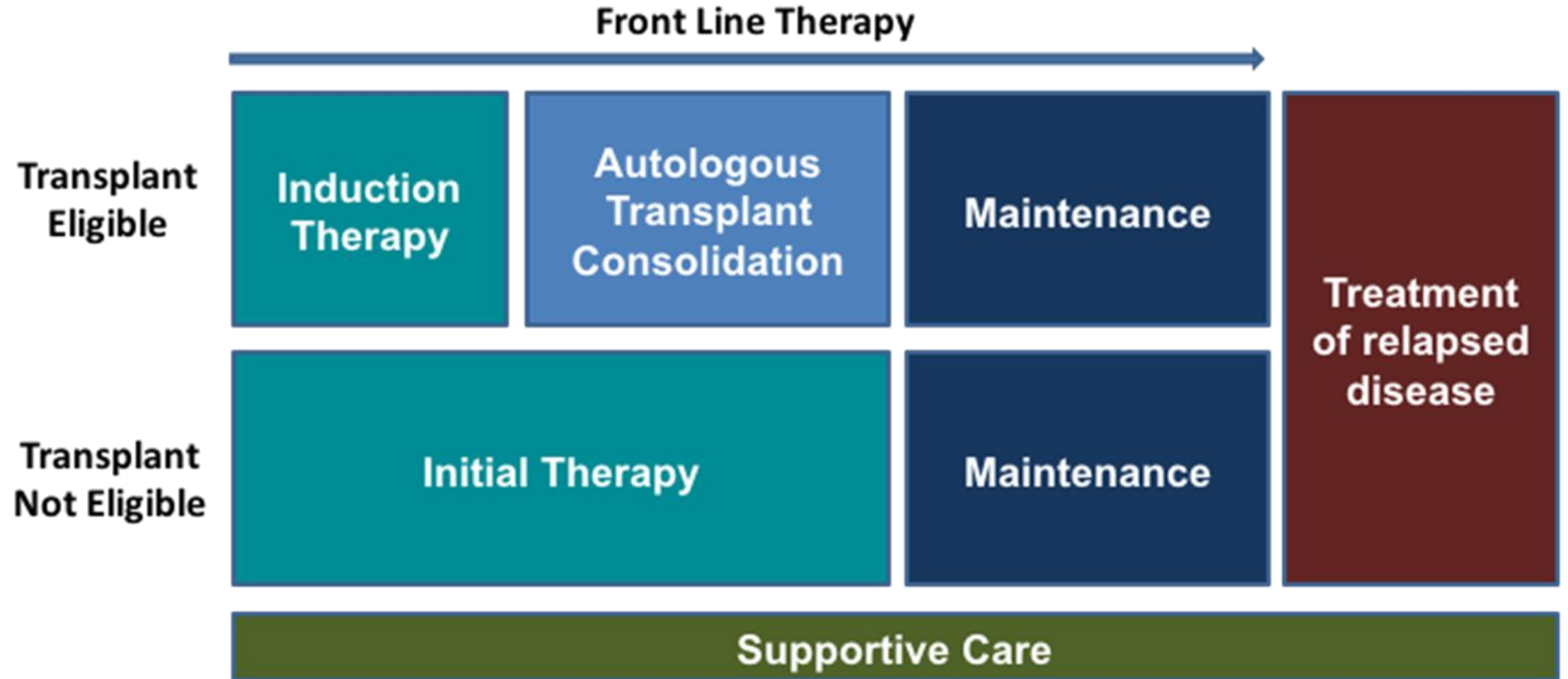
Consider frailty status when treating older patients (>80yo)

Treatment objectives and strategies may be different for fit and frail patients

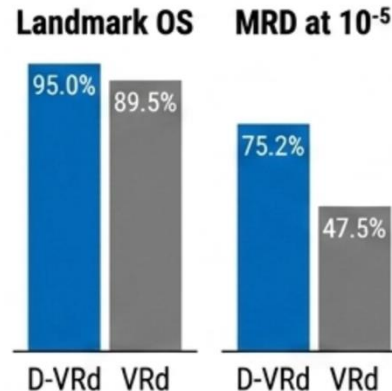
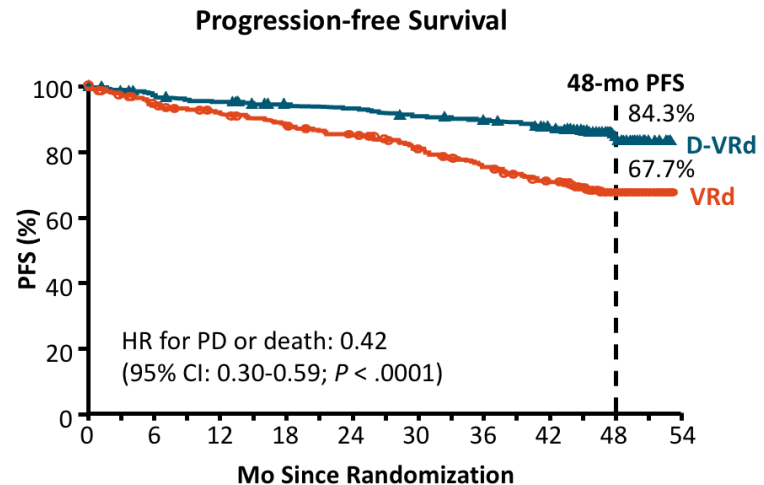
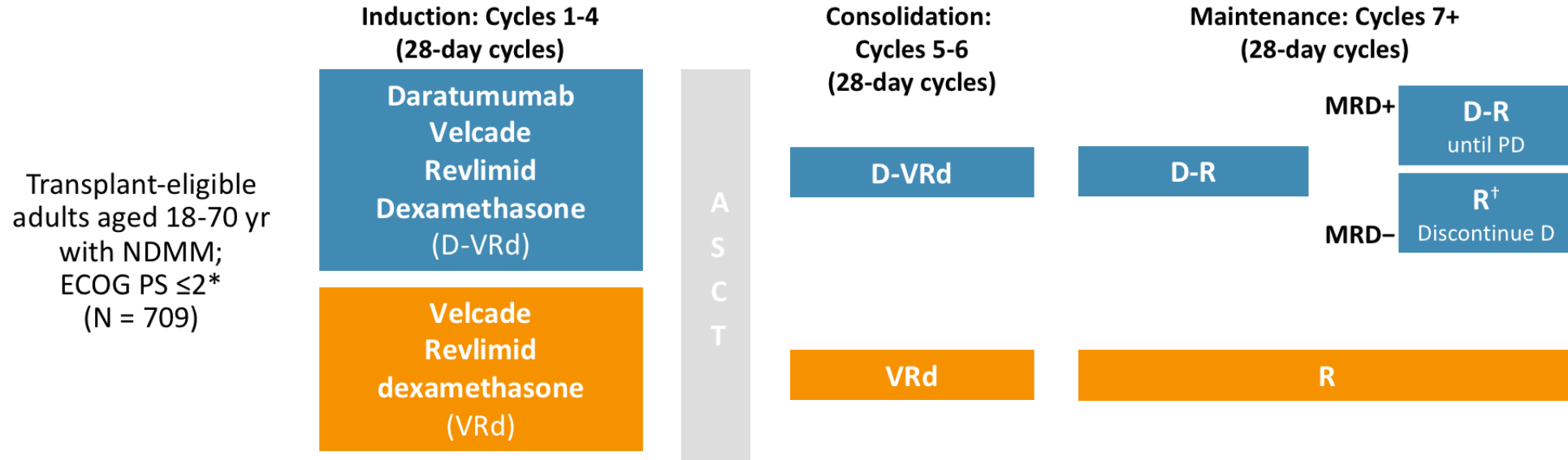
Improving MRD negativity rates for fit patients *to improve survival*

Limiting toxicity for frail patients, to maintain dosing, *not to worsen survival*

The Current Roadmap



SoC in Transplant Eligible MM: Dara



Toxicities



Cytopenias: High
(expected with quads)

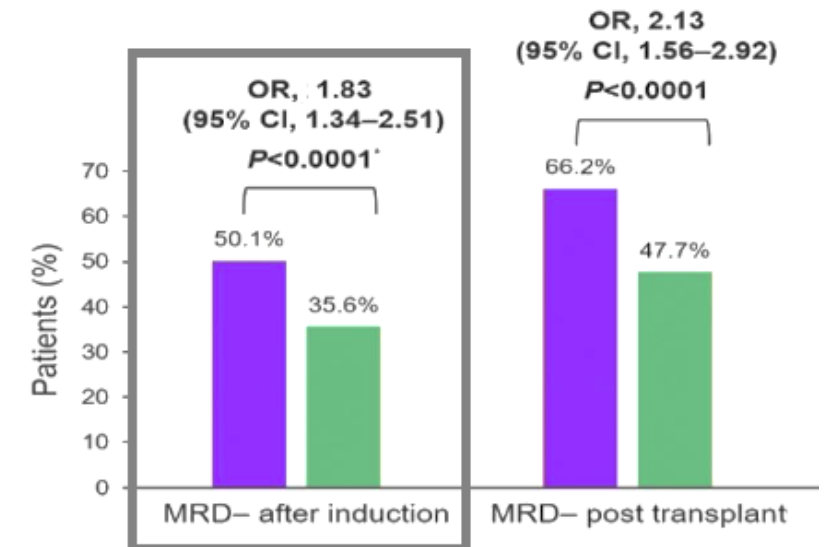
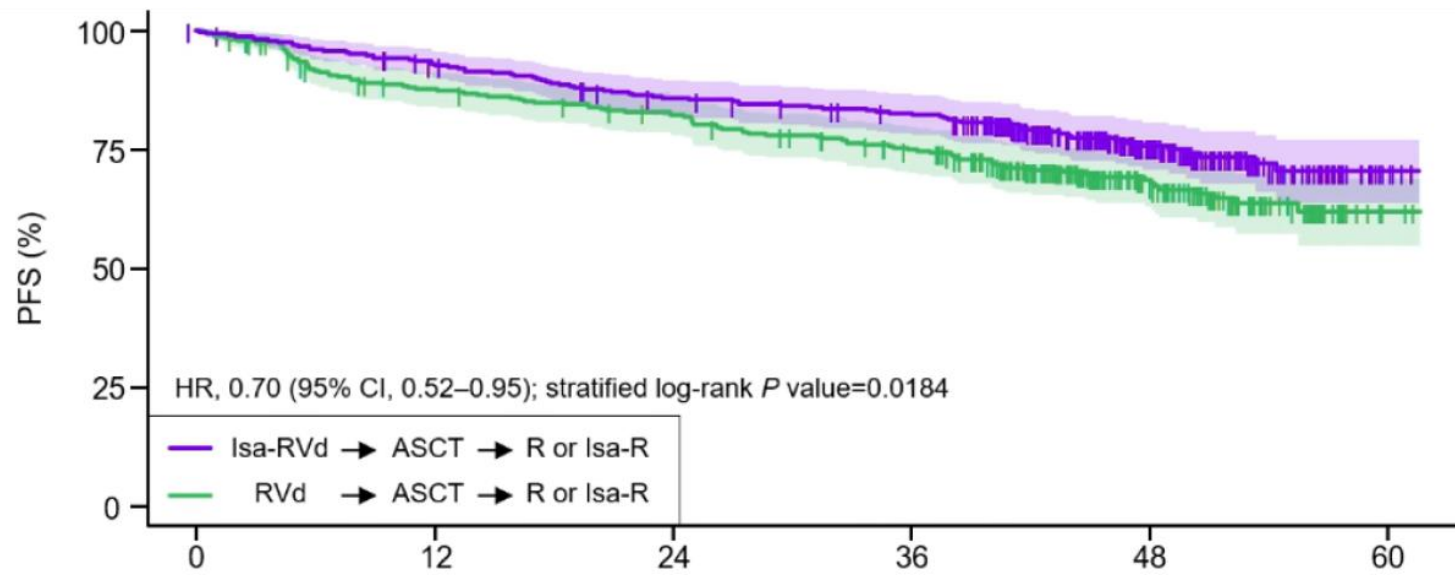
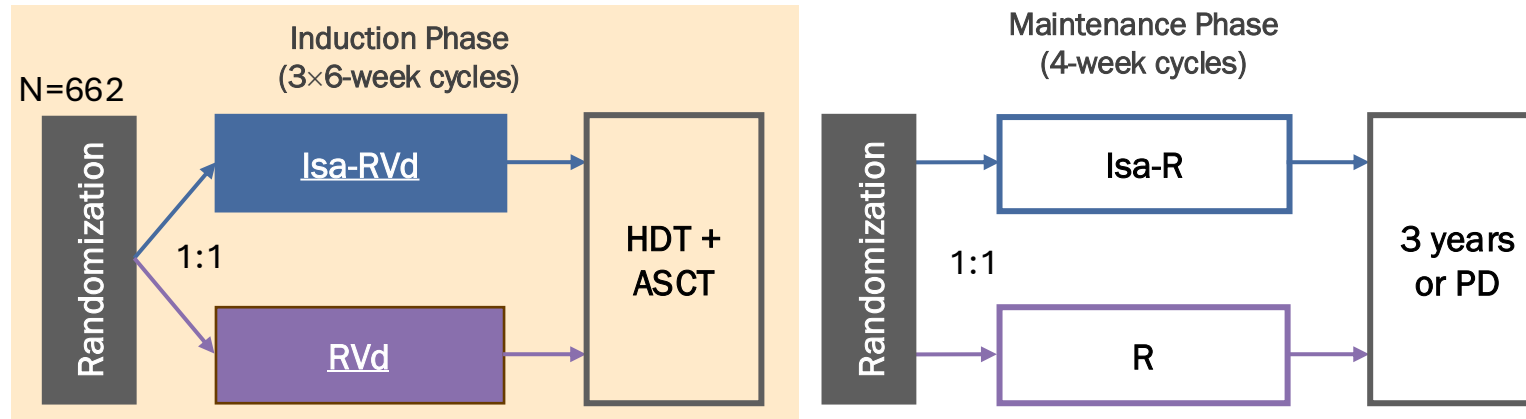


Infection:
Elevated but manageable



Neuropathy: Lower/Stable
(Subcutaneous administration mitigates risk)

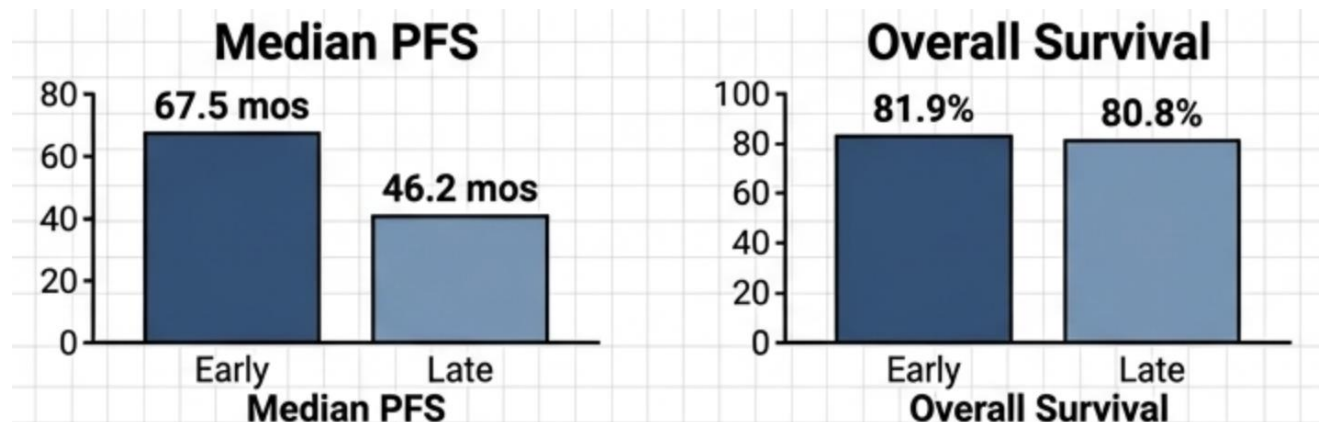
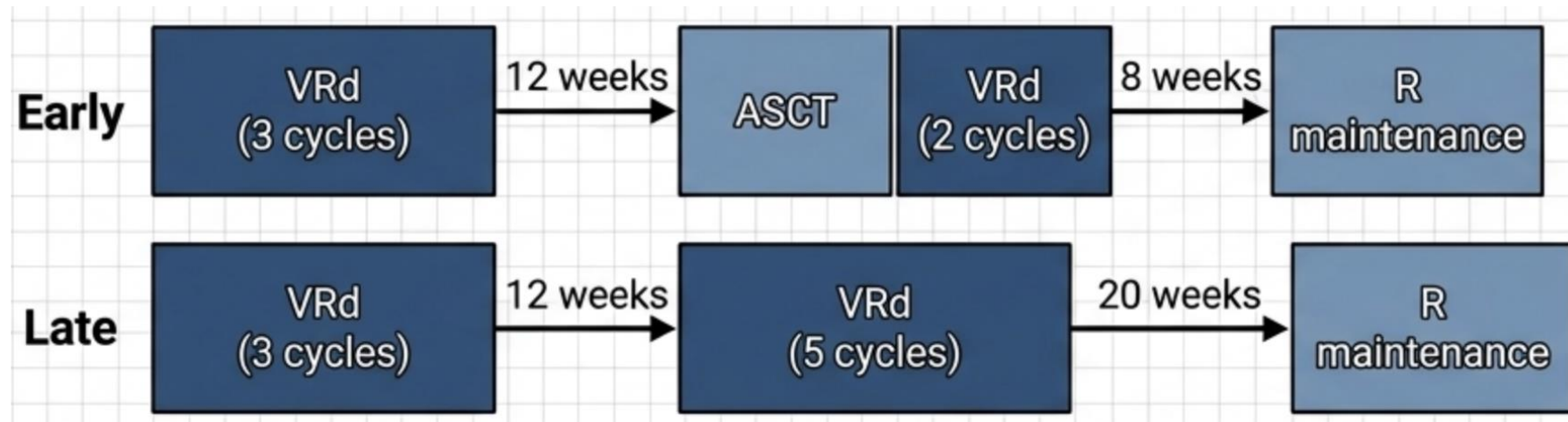
SoC in Transplant Eligible MM: Isa





Do I **HAVE** to Get a Transplant?

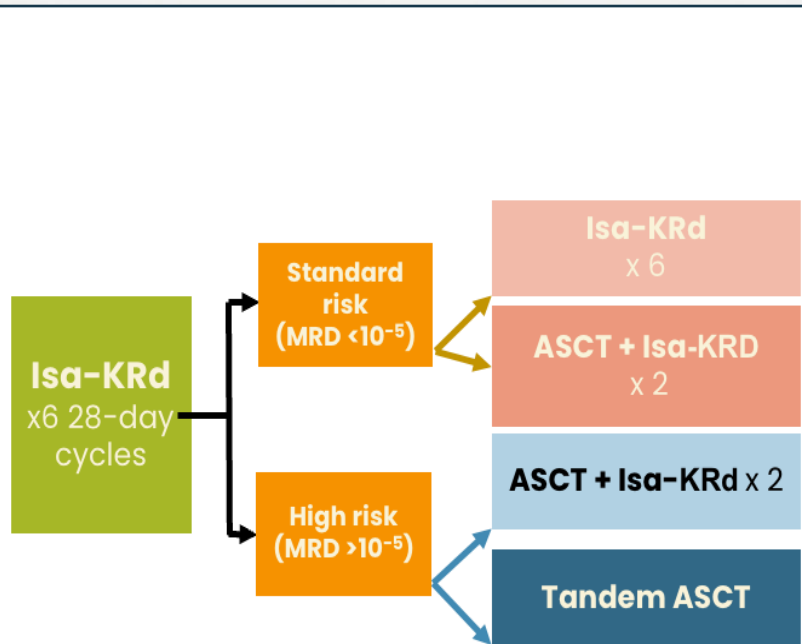
DETERMINATION: Early vs Late ASCT



Can a Test Help Me Decide When?

MIDAS: Measurable Residual Disease (MRD)-adapted

Schema



Results

Can MRD dictate the need for ASCT?

Outcome: MRD at 10⁻⁶

MRD- Group



(No significant difference)

MRD+ Group



(No significant difference)

Toxicities



Standard toxicity profiles



Standard toxicity profiles

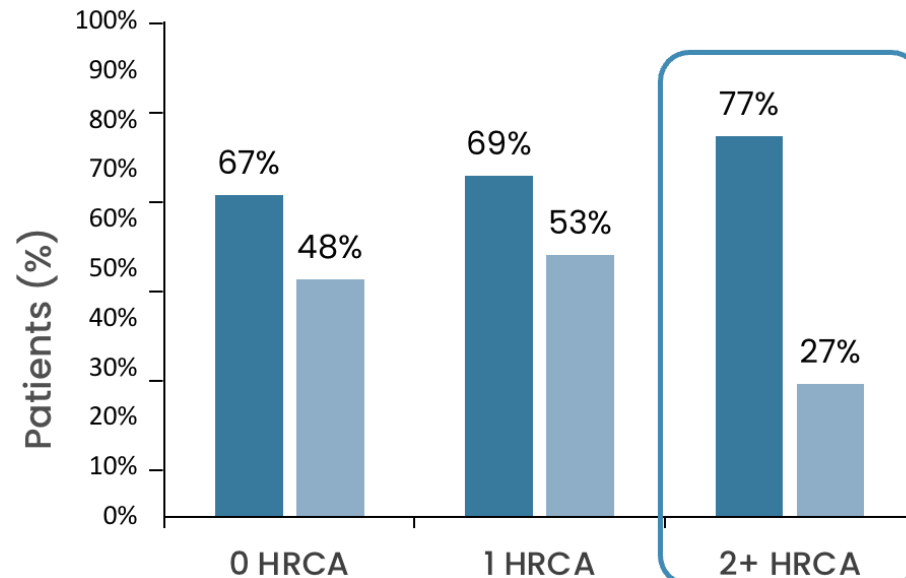
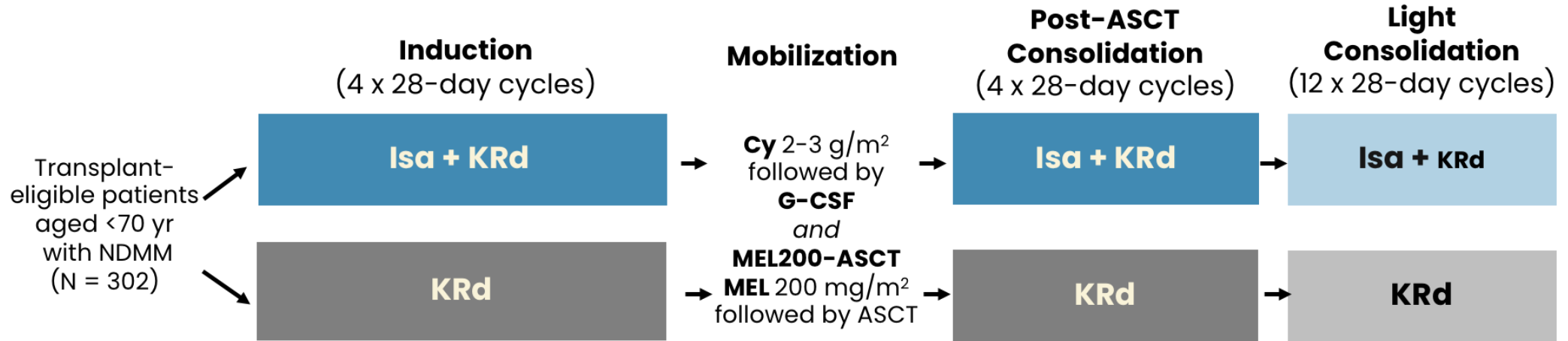


15% of tandem ASCT patients dropped out before 2nd.



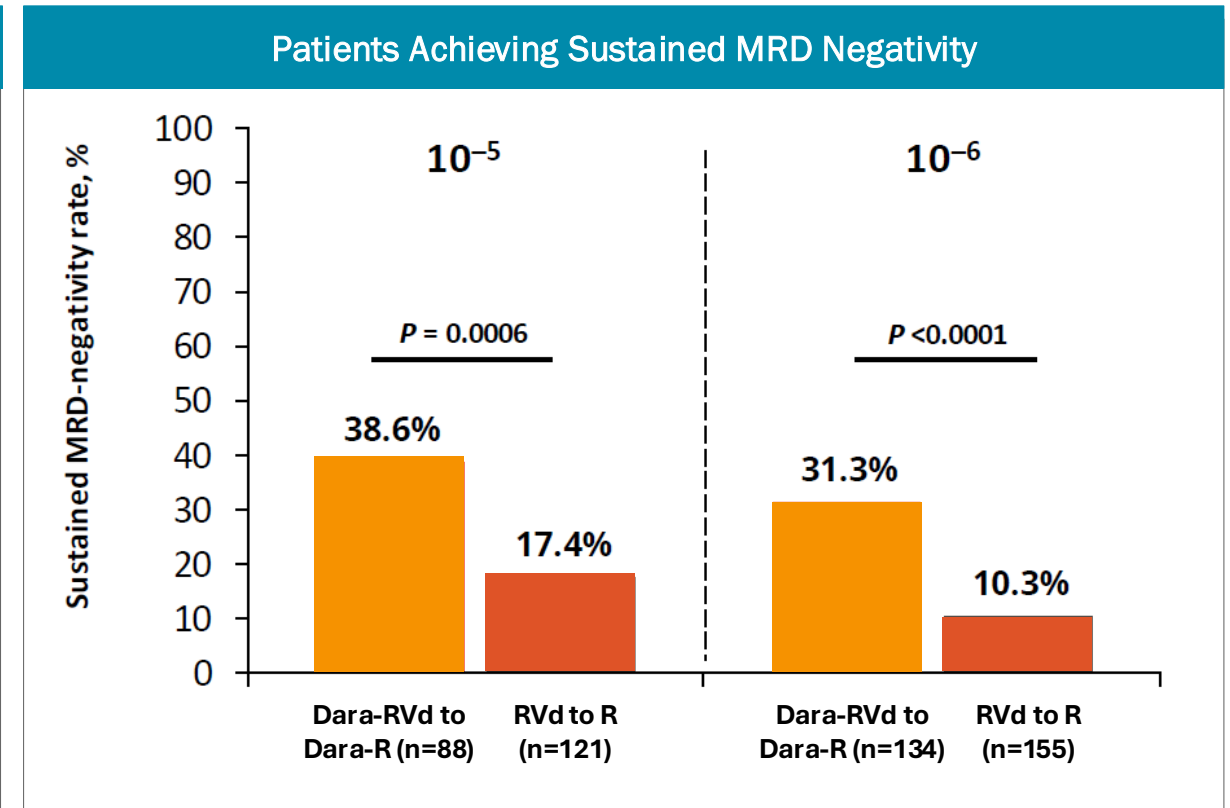
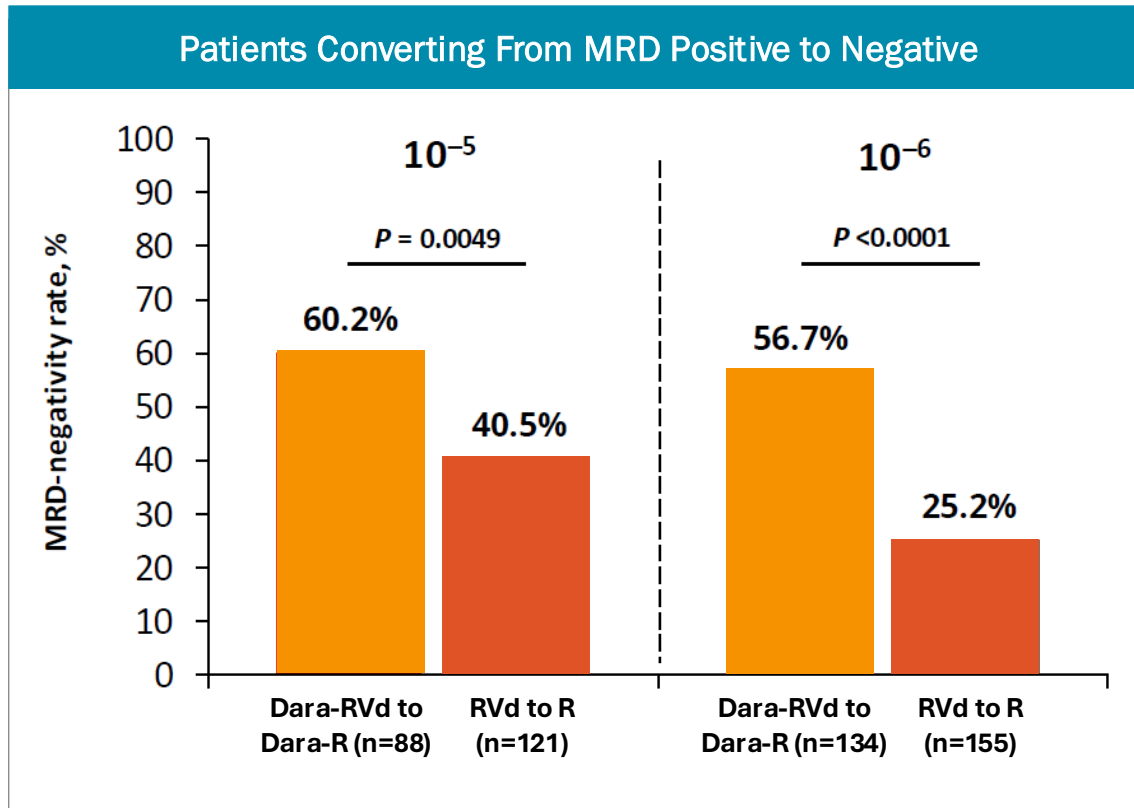
Cardiac Risk: Requires monitoring (Carfilzomib), though non-heme AEs similar across arms

What If I Have High-Risk MM?



High-risk cytogenetic abnormalities: del(17p13.1), t(4;14) (p16.3;q32.3), t(14;16) (q32.3;q23), gain(1q21), or amp(1q21)

We Haven't Talked About Maintenance!

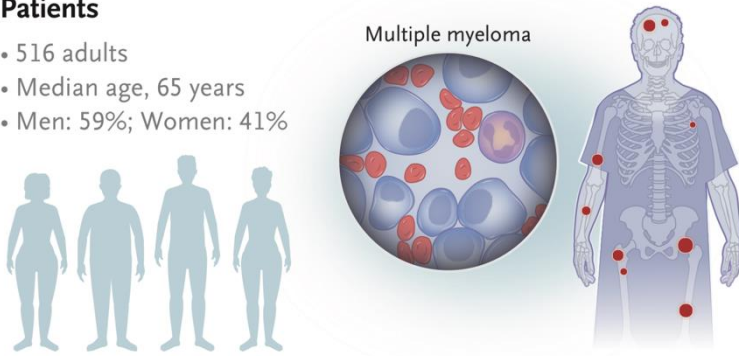


- With Dara-RVd + Dara-R maintenance vs RVd + R maintenance:
 - Rate of conversion to MRD negative at 10^{-6} more than doubled (57% vs 25%)
 - Conversion to sustained MRD negativity at 10^{-6} tripled (31% vs 10%)
 - **Pending results of the cohort that discontinued Daratumumab in the Dara-R arm....**

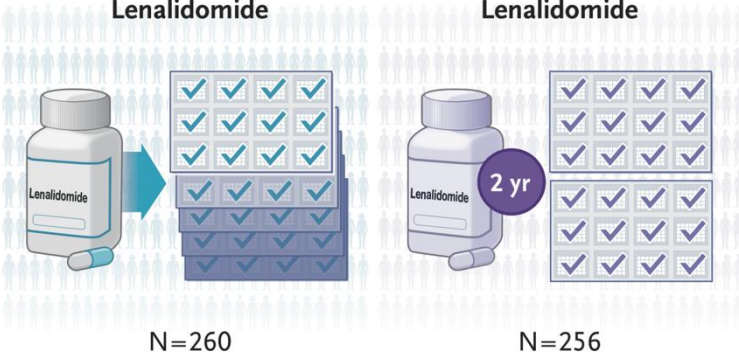
OK, But For How Long?

Patients

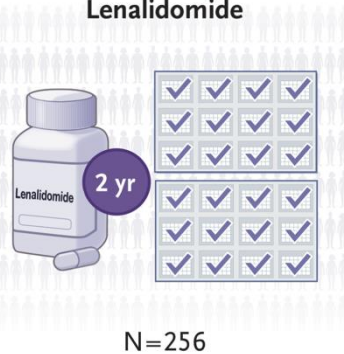
- 516 adults
- Median age, 65 years
- Men: 59%; Women: 41%



Indefinite-Duration Lenalidomide

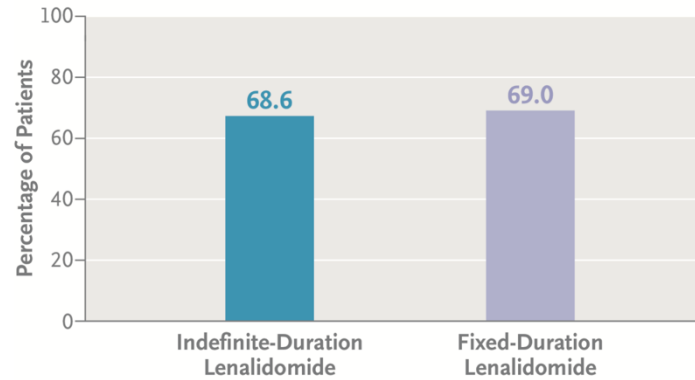


Fixed-Duration Lenalidomide

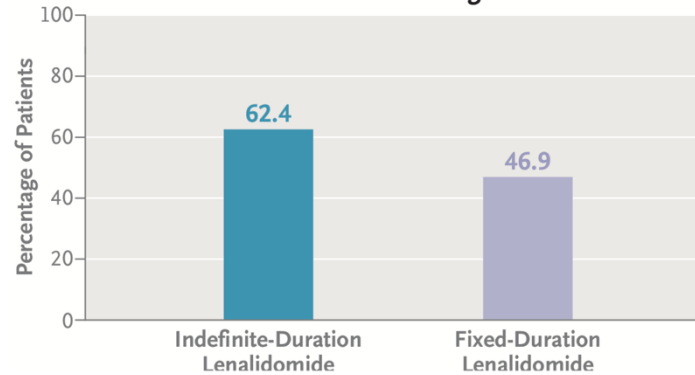


Overall Survival

Difference, -0.4 percentage points (95% CI, -9.0 to 8.3); P=0.93



Hematologic and Nonhematologic Adverse Events of Grade 3 or Higher



Toxicities



Cytopenias:
Higher in indefinite



Secondary Cancers:
11.2% (Indefinite) vs
8.3% (Fixed)



**Grade 3+ Non-heme
AEs:**
48.2% vs 31.5%

No new safety signals.

Can I Get Rid of Steroids Earlier?



Down With Dex!

Dex De-escalation Protocol

1. Assess 40 mg vs 20 mg weekly starting dose
2. Limit starting dose to 2-3 mo
3. Then taper dose each mo over the course of 3-4 mo
4. Plan to stop dexamethasone at 6 mo
5. Regular assessments for toxicity and de-escalate sooner if clinically indicated

CLINICAL TRIALS AND OBSERVATIONS

Comment on [Banerjee et al](#), page 75

Down with dex!

Susan Harding¹ and Joseph Mikhael^{1,2} | ¹HonorHealth Research Institute and ²Translational Genomics Research Institute

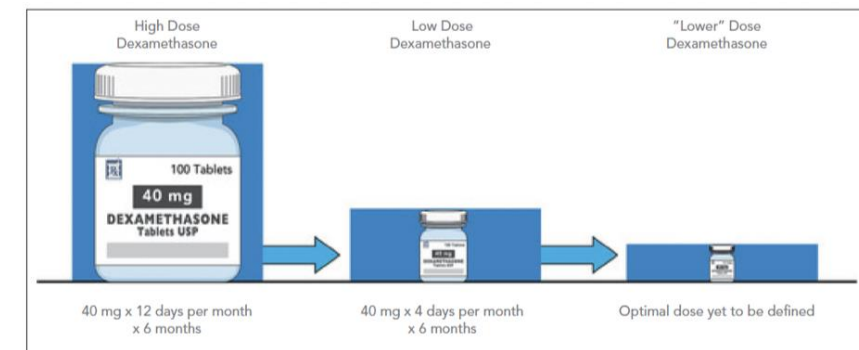
In this issue of *Blood*, Banerjee et al¹ present a secondary analysis of pooled data from 2 completed SWOG cooperative group trials (S0777 and S1211), showing that dose reductions in dexamethasone had no significant impact on postinduction progression-free survival (PFS) and overall survival (OS) in patients with newly diagnosed multiple myeloma.

Historically, with limited options for the treatment of multiple myeloma, dexamethasone was a necessary therapeutic partner to treat the disease.^{2,3} Dexamethasone doses used to invoke treatment responses in the past were as high as 40 mg for 4 days per week. Although dexamethasone demonstrates initial efficacy, especially at robust dosing, it also demonstrates significant and

problematic toxicity over time. When the ECOG E4A03 study demonstrated increased mortality at this high 4-day pulse dexamethasone dosing (40 mg given 12 of 28 days) compared with 40 mg once weekly,⁴ we saw our first big decrease in dexamethasone dosing in myeloma treatment regimens. The recent exponential increase in the number of effective myeloma therapies and

expanded use of quadruplet induction therapy brings into question the appropriate dose and duration of dexamethasone in contemporary myeloma treatment. In this exciting landscape of effective treatment options, it is unclear to what extent dexamethasone is necessary to treat disease. Dexamethasone is also used for symptom management and infusion reaction prophylaxis, although these supportive care indications generally do not require 40 mg.

With patients with multiple myeloma living longer than ever, it is important to minimize treatment toxicities. Providers and patients with myeloma are no strangers to the long list of bothersome steroid toxicities, including insomnia, heartburn, edema, psychiatric disturbances, muscle weakness, hyperglycemia, and diabetic complications. Too often, novel myeloma drugs are disrupted or abandoned due to symptoms that may, in fact, be dexamethasone induced. There is a need to better identify



Protecting Your Bones and Blood Vessels



Bones

bone-modifying agent + dental check



Blood vessels

clot-risk-matched prevention

Bone-strengthening medicine typically starts early, often after a dental check first (Zometa or Xgeva).

Blood-clot prevention (matched to your personal risk) usually starts alongside your myeloma treatment, especially with IMiDs (lenalidomide).

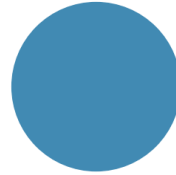


Preventing Infections



Antivirals

shingles prevention



Vaccines

personalized schedule



PJP prophylaxis

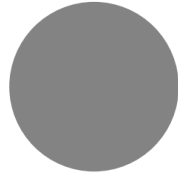
during high-dose steroids

Antivirals to prevent shingles is standard on a proteasome inhibitor or monoclonal antibodies.

Extended steroid use may require lung-infection (PJP) prevention. Antibiotics (levofloxacin) for the first 3 months can be beneficial.

Follow a personalized vaccination schedule, and sometimes immune-support infusions (IVIG) if antibody levels run low.

Protecting Your Kidneys and Nerves



Kidneys

dose-adjusted, avoid NSAIDs



Nerves

report tingling/numbness early

Doses are adjusted for kidney function; pain relievers like ibuprofen are avoided.

Report tingling/numbness in hands or feet right away — catching neuropathy early prevents permanent damage.

When to Make Sure a Specialist Is Involved



**High-risk /
double-hit**

**Not responding
to treatment**

**Aggressive /
out-of-marrow**

**Relapse
< 18 months**

**Transplant
evaluation**

Strong signals: high-risk/“double-hit” genetics, disease not responding to first treatment, an aggressive or out-of-marrow presentation, or being evaluated for transplant.

Relapse within ~18 months of finishing initial treatment/transplant is another key signal.

Can We Talk About the “C” Word?

Proposed Definition of Cure

- Patient could be newly-diagnosed or with relapsed myeloma
- Patient in sustained Complete Remission (CR) by standard criteria
- MRD negative at 10^{-6} sensitivity by either NGS or NGF
- MRD negativity Sustained for 5 years **WITHOUT** any therapy.
(Irrespective of length of previous sustained MRD negativity on therapy)
 - There should be at least 4 MRD assessments in 5 years, last one being at 5 years; all negative and no positive test in between, if performed.
- Negative by functional imaging (PET/CT or DWI WB MRI)
 - At least at the beginning of 5 years period and at the end of 5 years period with no intervening positivity, if performed.

Questions?



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